

Haagerup's Constructive Vector Need Not Be cbF-Optimal

An explicit rank-two projection counterexample to a question in arXiv:2312.09029

Automated mathematics run `fa_banach_001`, lane 14

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Status. Candidate full counterexample, likely valid, pending human review. The proof is exact; numerical optimization is used only as an independent sanity check.

1 Source question

The source is Erik Christensen, *Some points of view on Grothendieck's inequalities*, arXiv:2312.09029 [1]. The introduction asks whether Haagerup's method actually produces the optimal cbF-factorization (or the corresponding operator factorization) of a complex matrix. In Section 3, after proving optimality for matrices with nonnegative entries, the paper says that in the general case Haagerup's vector might remain optimal, but the author expects otherwise and has no definite result. The passage begins at the foot of p. 11 and continues on p. 12 of the arXiv PDF:

It seems natural to ask if the results above can be extended to matrices with real entries only? Since it is known that $k_G^{\mathbb{R}} > 1$, this is

12

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not possible, but why not. One reason, we anticipate, is that for some matrices with real entries the maximal value of $\|F_X(u)\|_2$ must be obtained in a unitary $u = (u_1, \dots, u_n)$ where the entries are not all real. In the general case, it might still be that Haagerup's vector continues to be the one from the optimal cbF-factorization, but we think that this is not the case. We do not have a definite result which proves this, but we have made some observations, which indicate this and also give some more information on the problems in finding the optimal vector ξ for the cbF-factorization.

Figure 1: Full-width crops from pp. 11–12 of the source PDF.

The relevant general-case sentence is:

“In the general case, it might still be that Haagerup’s vector continues to be the one from the optimal cbF-factorization, but we think that this is not the case. We do not have a definite result which proves this.”

We give the requested definite negative result.

2 Definitions and an equivalent diagonal optimization

For $z \in \mathbb{C}^n$, write $\Delta(z)$ for the diagonal matrix with diagonal z . For a complex $m \times n$ matrix X , the source’s factorization norm can be written as

$$\|X\|_{\text{cbF}} = \min_{\substack{\xi \in \mathbb{R}_+^n \\ \|\xi\|_2=1}} \|X\Delta(\xi)^{-1}\|_{\text{op}}, \quad (1)$$

with the usual support convention if a coordinate of ξ vanishes. In our example every feasible diagonal below is strictly positive, so no support subtlety occurs.

The ordinary F -norm is

$$\|X\|_F = \max_{|u_j|=1} \|Xu\|_2.$$

For a maximizing unimodular vector u , Haagerup’s construction assigns the probability vector

$$\lambda_j = \frac{\overline{u_j}(X^*Xu)_j}{\|X\|_F^2}, \quad \xi_{H,j} = \sqrt{\lambda_j}. \quad (2)$$

The maximizing property makes the λ_j real and nonnegative, and their sum is one; this is precisely the state construction in Section 3 of the source [1].

Lemma 1 (Diagonal SDP form). *For $P = X^*X$,*

$$\|X\|_{\text{cbF}}^2 = \min\{\text{Tr } D : D \text{ is diagonal and } D \succeq P\}. \quad (3)$$

If ξ is optimal in (1) and $t = \|X\|_{\text{cbF}}^2$, then $D = t\Delta(\xi)^2$ is optimal in (3). Conversely, an optimal D of trace t gives the optimal vector $\xi_j = \sqrt{D_{jj}/t}$.

Proof. For a positive unit vector ξ , put

$$t(\xi) = \|X\Delta(\xi)^{-1}\|_{\text{op}}^2 = \|\Delta(\xi)^{-1}P\Delta(\xi)^{-1}\|_{\text{op}}.$$

The inequality $\Delta(\xi)^{-1}P\Delta(\xi)^{-1} \preceq t(\xi)I$ is equivalent to $P \preceq t(\xi)\Delta(\xi)^2$. Thus $D = t(\xi)\Delta(\xi)^2$ is feasible and $\text{Tr } D = t(\xi)$. Conversely, if diagonal $D \succeq P$ has trace t , set $\xi_j = \sqrt{D_{jj}/t}$. Then $\|\xi\|_2 = 1$ and $P \preceq t\Delta(\xi)^2$, so $t(\xi) \leq t$. Taking minima in both directions proves (3) and the optimizer correspondence. $\square$

3 Proof intuition

We design a two-dimensional subspace $L \subset \mathbb{C}^4$ with two competing properties. First, L has a positive frame operator Q whose four diagonal entries are all one. This gives an exact dual witness forcing the unique optimal diagonal in (3) for the projection onto L to be I_4 ; hence the unique optimal factorization vector is uniform. Second, L has no unimodular vector. If Haagerup’s weights were uniform, the maximizing unimodular vector in (2) would have to be a nonzero eigenvector of the projection with eigenvalue in $(0, 1]$, forcing it into L . The two properties are incompatible, so Haagerup’s vector cannot be optimal.

4 The counterexample

Theorem 1. *Let*

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2^{-1/2} & 2^{-1/2} \\ 2^{-1/2} & i2^{-1/2} \end{pmatrix}, \quad P = A(A^*A)^{-1}A^*. \quad (4)$$

Then P is a 4×4 orthogonal projection of rank two. For $X = P$, the cbF-optimal positive unit vector is uniquely

$$\xi_0 = \frac{1}{2}(1, 1, 1, 1), \quad \|P\|_{\text{cbF}}^2 = 4.$$

For every maximizing unimodular vector u used in Haagerup's construction, the vector ξ_H from (2) differs from ξ_0 . Consequently Haagerup's cbF-factorization, and hence its corresponding operator factorization, is strictly nonoptimal for this explicit matrix.

Proof. The range of A is

$$L = \left\{ \left(a, b, \frac{a+b}{\sqrt{2}}, \frac{a+ib}{\sqrt{2}} \right) : a, b \in \mathbb{C} \right\}. \quad (5)$$

The two columns of A are independent, so (4) is the orthogonal projection onto L .

We first show that L contains no unimodular vector. If a vector in (5) had all four coordinates of modulus one, then $|a| = |b| = 1$. The third coordinate would give

$$|a+b|^2 = 2 \implies \Re(a\bar{b}) = 0,$$

while the fourth would give

$$|a+ib|^2 = 2 \implies \Im(a\bar{b}) = 0.$$

This is impossible because $|a\bar{b}| = 1$.

Now put $Q = AA^*$. Each row of A has Euclidean norm one, so

$$Q \succeq 0, \quad \text{diag}(Q) = (1, 1, 1, 1), \quad \text{ran } Q = L. \quad (6)$$

Since P is the projection onto L , we have $PQ = Q$, and therefore

$$\text{Tr}(PQ) = \text{Tr } Q = 4. \quad (7)$$

The diagonal $D = I_4$ is feasible in (3), because $P \preceq I_4$. For any other feasible diagonal $D \succeq P$, positivity and (6)–(7) give

$$0 \leq \text{Tr}((D - P)Q) = \sum_{j=1}^4 D_{jj}Q_{jj} - \text{Tr}(PQ) = \text{Tr } D - 4.$$

Thus the minimum in (3) is 4.

It is also unique. If equality holds, the positive matrices $D - P$ and Q satisfy $\text{Tr}((D - P)Q) = 0$. Hence $Q^{1/2}(D - P)Q^{1/2} = 0$, which implies that $D - P$ annihilates $\text{ran } Q = L$. On L the projection P is the identity, so $Dv = v$ for every $v \in L$. Every coordinate is nonzero on some vector of L ; since D is diagonal, this forces $D_{jj} = 1$ for all j . Therefore $D = I_4$ is the unique optimizer. By the diagonal SDP lemma, the unique optimal factorization vector is $\xi_0 = (1/2, 1/2, 1/2, 1/2)$.

It remains to rule out this vector for Haagerup's construction. Let u be any unimodular maximizer for $\|P\|_{\text{F}}$. Since P is an orthogonal projection,

$$\|P\|_{\text{F}}^2 = \|Pu\|_2^2 = u^*Pu < \|u\|_2^2 = 4. \quad (8)$$

The inequality is strict because equality for a projection holds exactly when $u \in L$, while L contains no unimodular vector.

Suppose for contradiction that Haagerup's vector were cbF-optimal. Uniqueness would give $\xi_H = \xi_0$, hence $\lambda_j = 1/4$ for all j . Formula (2), using $X^*X = P$, would then yield

$$(Pu)_j = \frac{\|P\|_F^2}{4} u_j \quad (1 \leq j \leq 4), \quad \text{or} \quad Pu = \frac{\|P\|_F^2}{4} u.$$

The scalar is positive, and the only nonzero eigenvalue of a projection is 1. Therefore $\|P\|_F^2 = 4$ and $u \in L$, contradicting (8). Thus no choice of maximizing u makes Haagerup's vector optimal. $\square$

5 Verification

The included script `code/verify_projection.py` performs exact SymPy checks of

$$P = P^* = P^2, \quad \text{rank } P = \text{rank } Q = 2, \quad \text{diag } Q = (1, 1, 1, 1), \quad PQ = Q, \quad \text{Tr}(PQ) = 4.$$

It also carries out a deterministic multistart phase optimization as a sanity check. Numerically it finds

$$\|P\|_F^2 \approx 3.930567311, \quad \lambda \approx (0.216773, 0.216773, 0.283227, 0.283227),$$

and the squared operator norm of the resulting Haagerup factor is about 4.367920, strictly above the exact optimum 4. These decimals are not used in the proof.

6 Novelty search, limitations, and review recommendation

A bounded web and local-corpus search was performed on 21 July 2026 using the exact question phrase and combinations of `Haagerup vector`, `optimal cbF-factorization`, the paper title, its author, and arXiv:2312.09029. The search found the source/published article and unrelated Grothendieck material, but no later answer and no matching projection construction. This is not an exhaustive priority search; novelty confidence is therefore moderate.

The theorem settles the source's general cbF/operator-factorization optimality question negatively. It does not attempt to optimize the dimension of a counterexample or classify matrices for which Haagerup's construction is optimal.

For human review, the most consequential checkpoints are:

1. the equivalence between the source's cbF factorization and (3);
2. the equality case in the witness argument with $Q = AA^*$; and
3. the use of uniform Haagerup weights to force a unimodular eigenvector of P .

References

- [1] E. Christensen, *Some points of view on Grothendieck's inequalities*, Linear Algebra Appl. 689 (2024), 22–42; arXiv:2312.09029.